



A+ LAB SERIES V5

Lab 1A: Examining PC Hardware

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Material in this Lab Aligns to the Following	
CompTIA A+ (220-1201) Exam Objectives	3.5: Given a scenario, install and configure motherboards, central processing units (CPUs), and add-on cards 5.2: Given a scenario, troubleshoot motherboards, RAM, CPUs, and power.
CompTIA A+ (220-1202) Exam Objectives	1.3: Compare and contrast basic features of Microsoft Windows editions. 3.1: Given a scenario, troubleshoot common Windows OS issues.

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Introduction

Hardware refers to the physical components of a computer system. These components include input devices (e.g., keyboard, mouse, touchscreen), output devices (e.g., monitor, printer, speakers), processing components (e.g., CPU, motherboard, RAM, GPU), storage devices (e.g., HDD, SSD, optical drive), and network components (e.g., NIC, modem, router).

Hardware requires software to function effectively. This software can include the operating system, which manages the computer's resources and interacts with hardware; applications, which are designed to perform specific tasks such as word processing, web browsing, or gaming; and drivers, which enable hardware devices to communicate with the operating system. For a computer system to operate optimally, all hardware and software components must be compatible. This means that they are designed to work together and support each other's functions.

Drivers are specialized software programs that enable these hardware devices to communicate effectively with the operating system.

Understanding computer hardware components and their configuration is essential for computer technicians. Careful consideration must be given to hardware choices to ensure compatibility, performance, and integration with existing systems. This includes knowing how to gather information about hardware, configure and manage the system when adding new components, and verify compatibility with software.

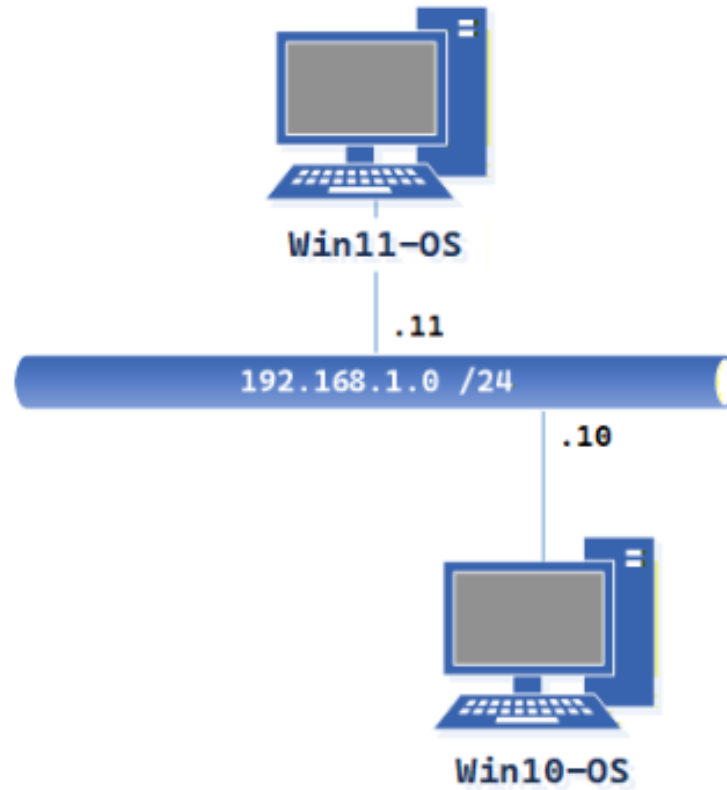
BIOS (Basic Input/Output System), while considered legacy, is a fundamental component of many older computer systems. It's the initial set of instructions executed by a computer upon startup, providing the framework for the CPU to interact with hardware devices. The BIOS/CMOS setup program allows users to modify BIOS settings, such as boot order and system parameters. These settings were traditionally stored in CMOS RAM but are now often stored on EEPROMs.

In modern computers, *UEFI (Unified Extensible Firmware Interface)* has largely replaced *BIOS*. *UEFI* offers several significant enhancements, including faster boot times, improved security, support for larger storage devices, a graphical user interface, and network boot support. While *UEFI* is backward compatible with *BIOS*, it offers a more modern and feature-rich platform for computer systems. Its advantages in terms of speed, security, and compatibility make it the preferred choice for most new systems.

Objective

In this lab, you will examine computer system hardware using *UEFI*, the *CMOS* utility, and the *Windows Device Manager*. Additionally, you will have the opportunity to review key concepts related to computer hardware, firmware, and system utilities.

Lab Topology



Lab Settings

The information in the table below will be needed in order to complete the lab. The task sections below provide details on the use of this information.

Virtual Machine	IP Address	Account (if needed)	Password (if needed)
Win10-OS	192.168.1.10 / 24	sysadmin	NDGLabpass123!
Win11-OS	192.168.1.11 / 24	sysadmin	NDGLabpass123!

1 Examine the UEFI/BIOS Configuration Using the Firmware Interface

UEFI and EEPROMs have largely replaced *BIOS and CMOS* in most modern computers; however, the terms are sometimes used interchangeably. BIOS, a legacy firmware, instructs the computer on essential functions and uses data stored in CMOS RAM to recognize and test hardware devices. UEFI, a more modern replacement, performs the same tasks but offers additional features and uses data stored in EEPROMs. Both BIOS and UEFI are crucial for the proper operation of a computer system.

The primary functions of BIOS include:

- Testing the mechanical operation of hardware
- Locating and launching the operating system
- Providing access to hardware devices for interaction with the operating system

Additionally, UEFI offers:

- Support for larger hard drives
- Faster boot times
- Increased security features

The specific BIOS setup program used depends on the motherboard manufacturer and BIOS version, not the operating system. To access the BIOS setup program, consult the motherboard documentation or look for instructions on the computer's splash screen during boot. The method for accessing the setup utility may vary among manufacturers.

1.1 Accessing the UEFI Utility in Windows 10

**Please
Note**

The steps in the lab only apply if your PC's motherboard has a *UEFI* firmware chip instead of a *BIOS* chipset and if *Windows 10* was installed using *UEFI*, and not legacy *BIOS*. These labs are being completed on virtual machines with legacy *BIOS* and as such do not have all the options available within the menus to complete all steps to make changes to *UEFI*, however it does provide you with practice in taking the steps to reach the area where the menu option for changing the *UEFI* setting from within *Windows 10* resides with *UEFI* configured. It is also a reminder of how to access the command line, and other troubleshooting tools from within *Windows 10*.

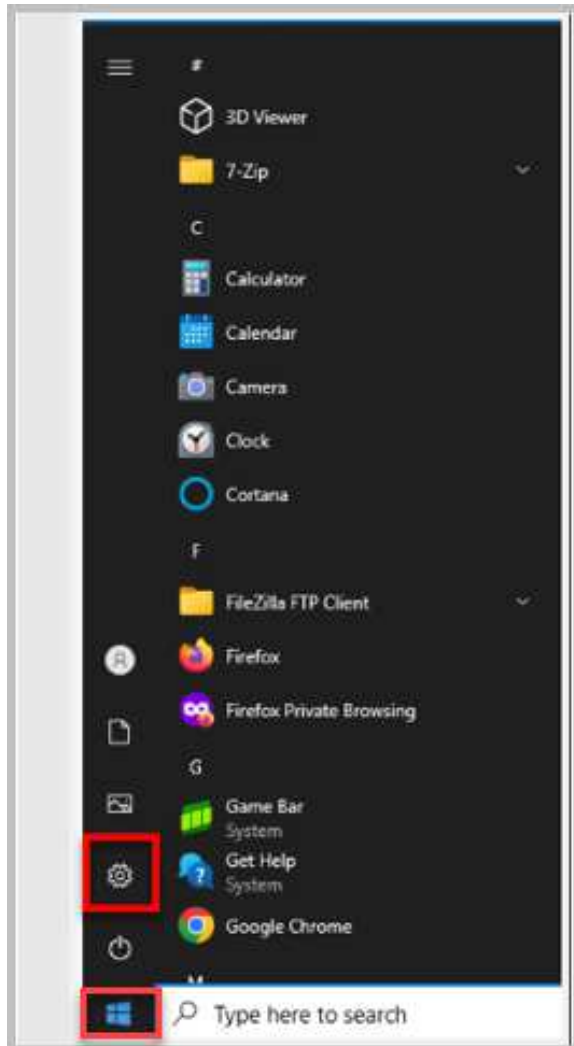
1. Open a console connection to the **Win10-OS** system.



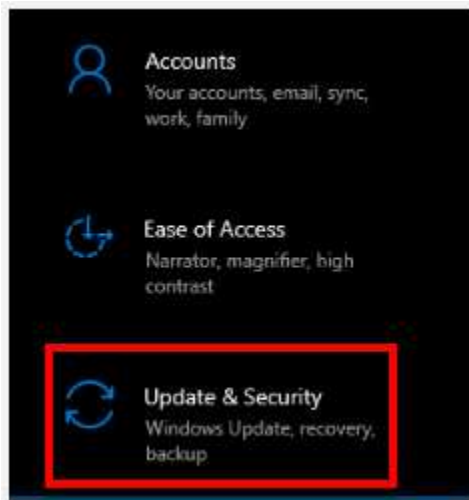


To launch the console for a lab device, you can click on the respective tab in the navigation bar at the top or navigate to the *Topology* section and click the lab device's corresponding image.

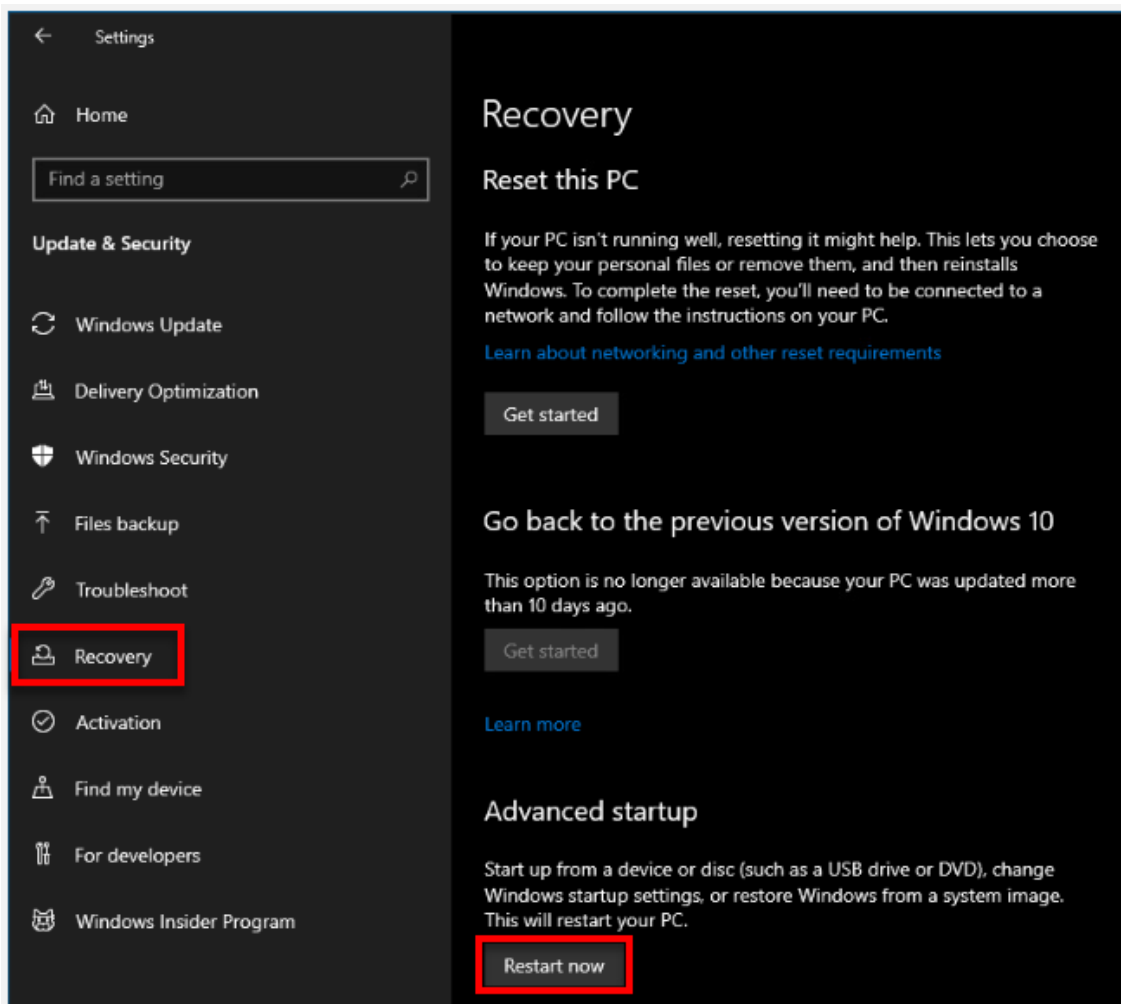
2. Click on the **Start** button and then click on the **Settings** icon.



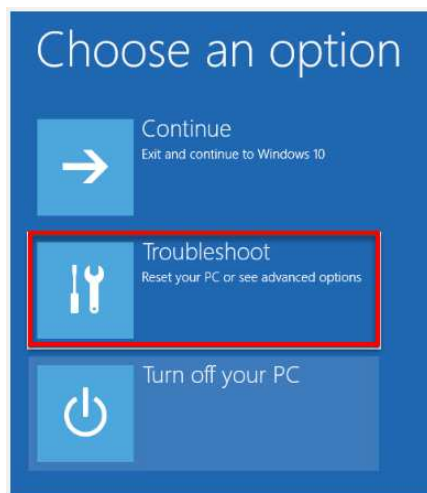
3. In the *Windows Settings*, click on **Update & Security**.



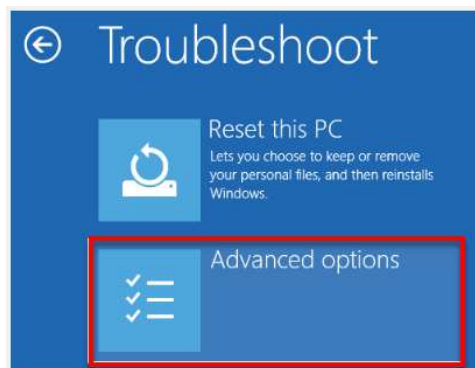
4. Click on **Recovery** from the left pane. Underneath the *Advanced Startup* heading, click on **Restart now**.



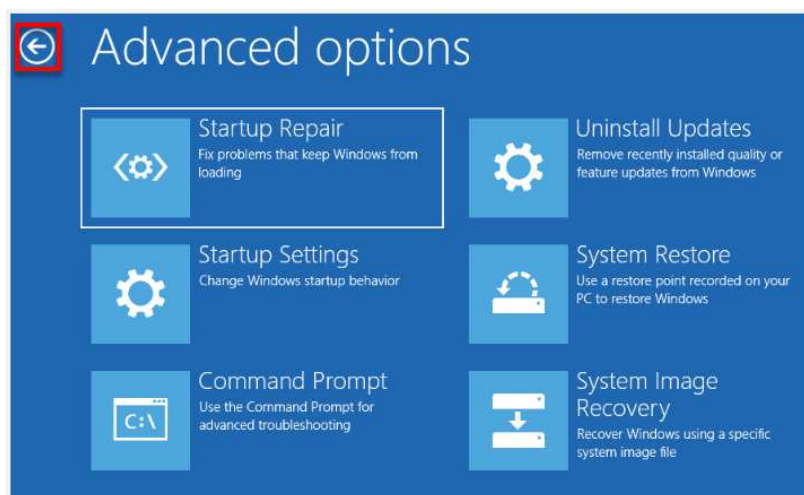
5. Once the system restarts on the *Choose an option* page, click on **Troubleshoot**.



6. Click on **Advanced Options**.



7. On the *Advanced Options* page, view the available options. If *UEFI* is configured, the *UEFI Firmware Settings* tile will be available. However, in this lab, the virtual machine does not support *UEFI*, and so the option is not presented here. When done viewing the options, click the **back arrow**.





If a machine supports *UEFI*, then the *UEFI Firmware Settings* option would be available. A screenshot below presents an example.



Once the *UEFI Firmware Settings* tile option is selected, a restart option becomes available as the next step so that the system can reboot to open the *UEFI* settings. As with the *BIOS*, these settings and interfaces can vary based on platform and manufacturer.

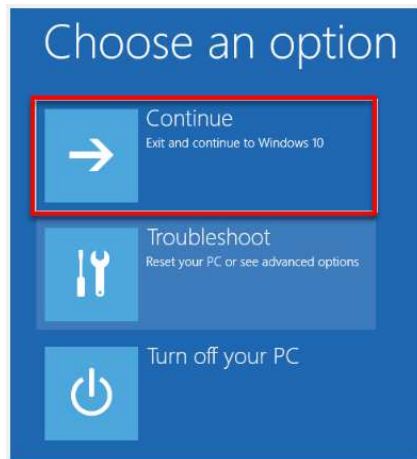
Accessing and configuring the *BIOS* will vary depending on the manufacturer, but common configurations are setting supervisor passwords, changing the boot order, and troubleshooting hardware. The system used to access the *BIOS* in this exercise was a *Win10* machine from a previous lab, but it is possible to do this on any system in the topology.

If you need to access low-level *UEFI* settings, there may be a slight difference. You may need to access the *UEFI* settings screen through the Windows boot options menu or the command line rather than pressing a key while your computer starts. Some PC systems with *UEFI* do allow you to access the *BIOS* in the same way, by pressing a key during the boot-up process. *UEFI* is platform independent, but with the fast boot of many platforms, including Windows 10, it is good to be aware of options other than trying to intercept the boot with key sequences.

8. Click the **back arrow** on the *Troubleshoot* page.



- Click the **Continue** button on the *Choose an option* page to normally boot to the operating system.



1.2 Accessing the UEFI Utility in Windows 11

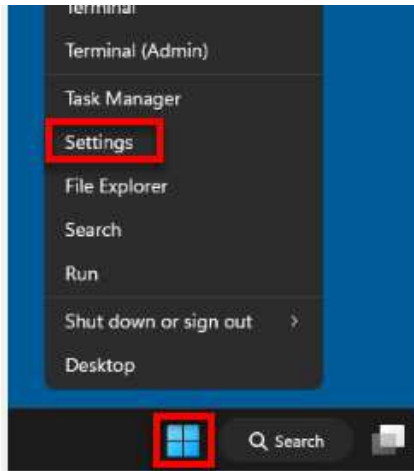
Please Note

The steps in the lab only apply if your PC's motherboard has a *UEFI* firmware chip instead of a *BIOS* chipset and if *Windows 11* was installed using *UEFI*, and not legacy *BIOS*. These labs are being completed on virtual machines with legacy *BIOS* and as such do not have all the options available within the menus to complete all steps to make changes to *UEFI*, however it does provide you with practice in taking the steps to reach the area where the menu option for changing the *UEFI* setting from within *Windows 11* resides with *UEFI* configured. It is also a reminder of how to access the command line, and other troubleshooting tools from within *Windows 11*.

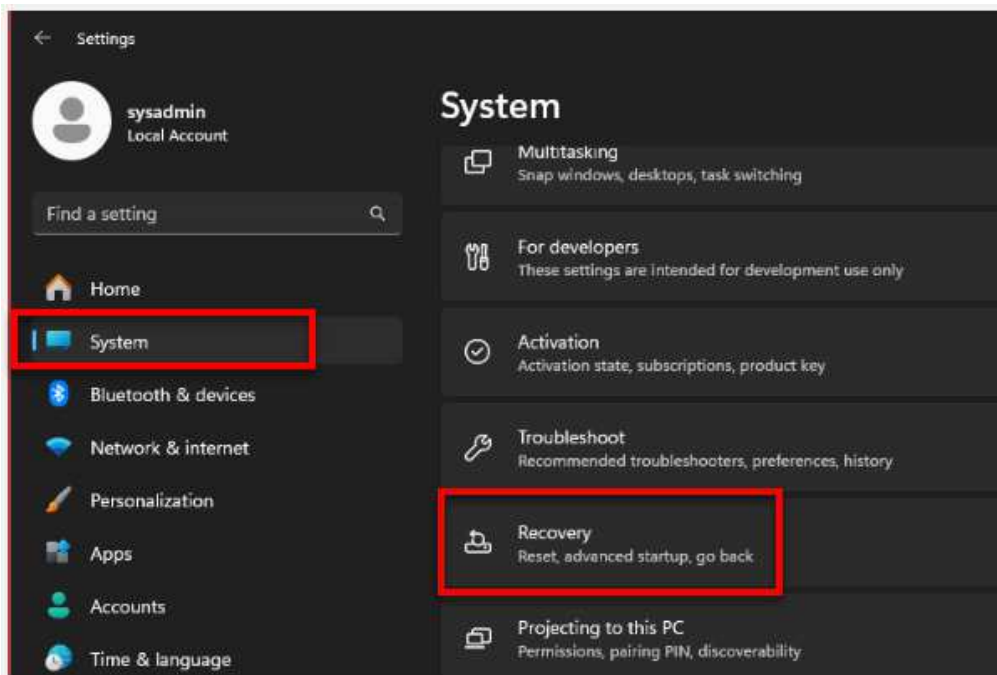
- Open a console connection to the **Win11-OS** system.



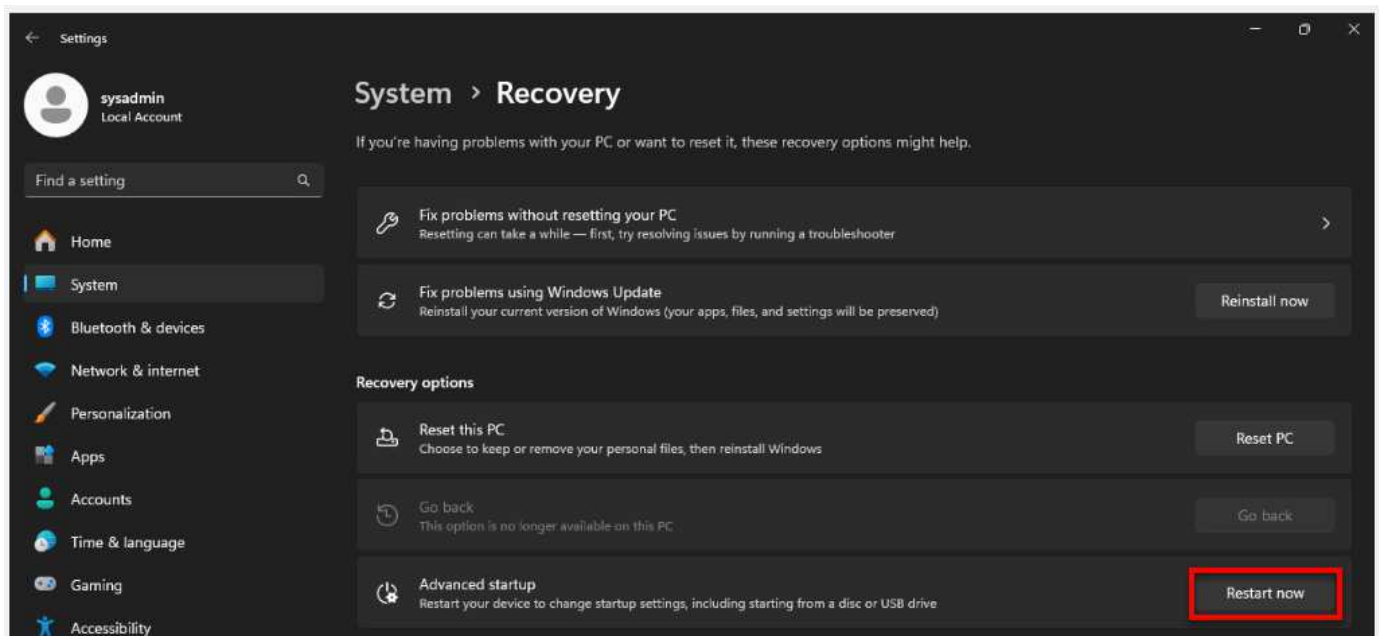
2. Right-click the **Start** button and then click on the **Settings** icon.



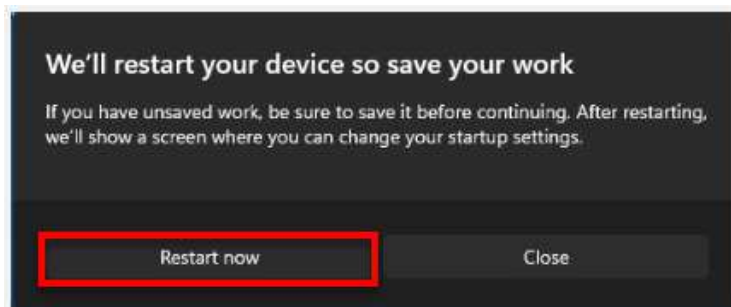
3. Select **System**. Next, click **Recovery** on the right side of the *System Page*. You will need to scroll through the system options to find the Recovery Link.



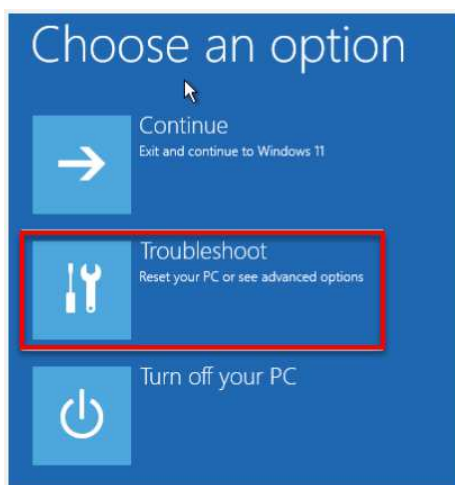
4. Click the **Restart now** button from the "Advanced startup" setting.



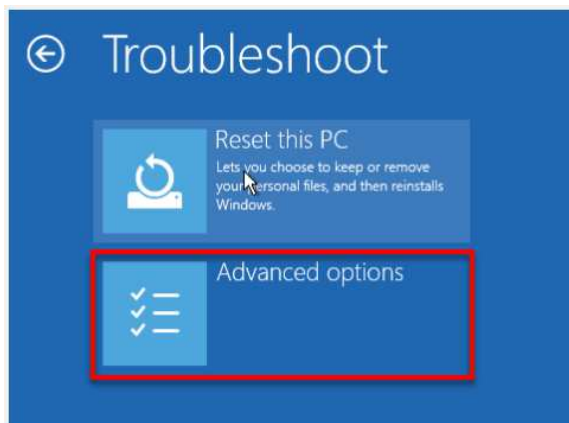
5. Click the **Restart now** button again.



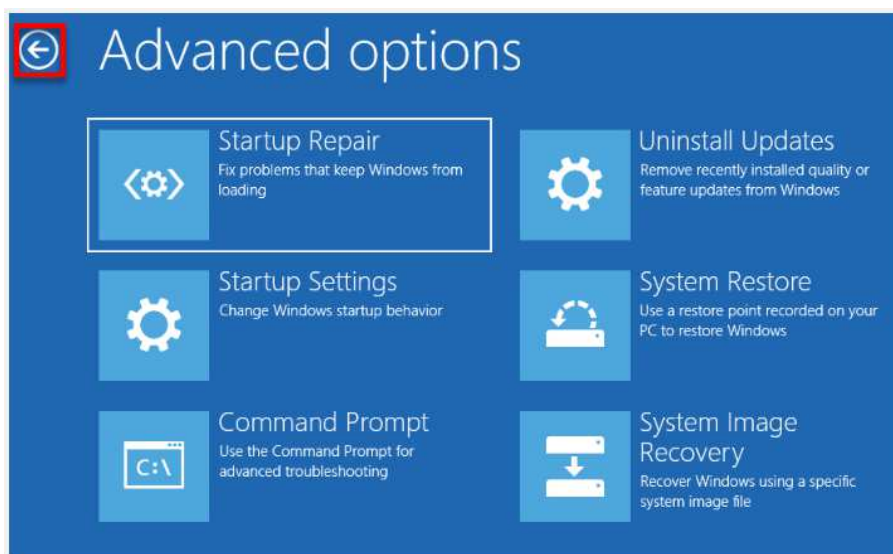
6. Click on **Troubleshoot** from the Windows Recovery Environment.



7. Click on **Advanced options**.



8. On the *Advanced Options* page, view the available options. If UEFI is configured, the *UEFI Firmware Settings* tile will be available. However, in this lab, the virtual machine does not support *UEFI*, and so the option is not presented here. When done viewing the options, click the **back arrow**.





If a machine supports *UEFI*, then the *UEFI Firmware Settings* option would be available. A screenshot below presents an example.



Once the *UEFI Firmware Settings* tile option is selected, a restart option becomes available as the next step so that the system can reboot to open the *UEFI* settings. As with the *BIOS*, these settings and interfaces can vary based on platform and manufacturer.

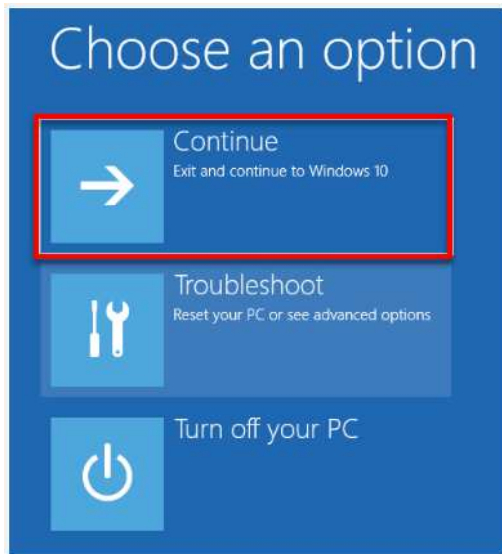
Accessing and configuring the *BIOS* will vary depending on the manufacturer, but common configurations are setting supervisor passwords, changing the boot order, and troubleshooting hardware. The system used to access the *BIOS* in this exercise was a *Win11* machine, but it is possible to do this on any system in the topology.

If you need to access low-level *UEFI* settings, there may be a slight difference. You may need to access the *UEFI* settings screen through the Windows boot options menu or the command-line rather than pressing a key while your computer starts. Some PC systems with *UEFI* do allow you to access the *BIOS* in the same way, by pressing a key during the boot-up process. *UEFI* is platform independent, but with the fast boot of many platforms, including Windows 10, it is good to be aware of options other than trying to intercept the boot with key sequences.

9. Click the **back arrow** on the *Troubleshoot* page.



10. Click the **Continue** button on the *Choose an option* page to normally boot to the operating system.



2 Using Device Manager

A PC technician needs to have a working knowledge of the utilities used to support and diagnose computer systems. There are many preloaded tools available in the *Windows* operating system. These tools allow users to manage, configure, and troubleshoot hardware and software on a computer without downloading third-party utilities.

Device Manager is a critical system utility in Windows 10 and 11 that provides a comprehensive overview of all hardware components installed on a computer. It serves as a central hub for managing, configuring, and troubleshooting devices, including printers, scanners, network adapters, and storage drives. The Device Manager categorizes devices into logical groups, such as "Computer," "Network adapters," "Disk drives," "Human Interface Devices," and more. Each device category can be expanded to reveal specific devices, their drivers, and their current status.



Device Manager is available in almost all versions of the *Windows* operating system, with slight differences in some versions. Working in *Device Manager* is the same in *Windows* (10/11). While there might be minor visual differences or interface tweaks, the core functionality and purpose remain consistent across both operating systems

Users can utilize the Device Manager to perform a variety of tasks, including:

- Updating drivers: Ensuring that devices are using the latest drivers for optimal performance and compatibility.
- Disabling or enabling devices: Temporarily turning off or on devices as needed.
- Diagnosing hardware conflicts: Identifying and resolving issues that may arise when devices compete for resources.
- Viewing device properties: Accessing detailed information about specific devices, such as their manufacturer, model number, and driver version.
- Managing power settings: Configuring power management options for devices to save energy or improve performance.

By providing a centralized interface for managing hardware, the Device Manager simplifies maintaining a computer's optimal performance and resolving hardware-related issues.

2.1 Accessing Device Manager from Control Panel to Examine Hardware



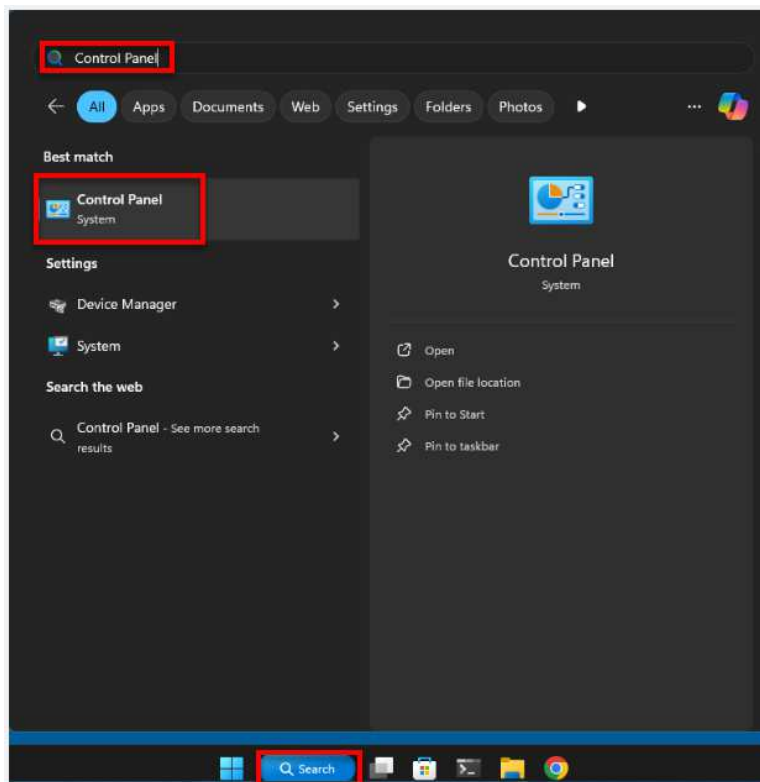
In Windows 11, the Control Panel is less prominently displayed and can be found by searching for it in the Start menu. Windows 10 places the Control Panel more prominently on the Start menu.

While the Control Panel in both Windows 10 and Windows 11 offers similar functionalities, Windows 11 has gradually incorporated more settings into the Settings app, making it a more comprehensive tool for managing system configurations.

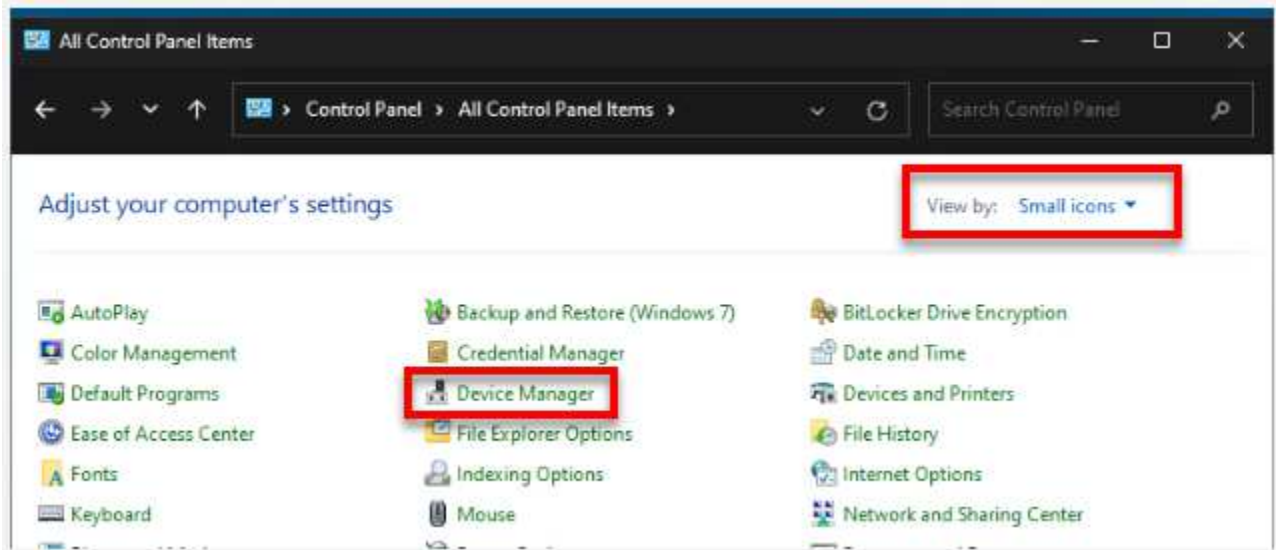
The Control Panel in Windows 10 has a more traditional look and feel than the redesigned Settings app in Windows 11.

There are multiple ways to open the *Control Panel* in *Windows 10* and *Windows 11*. Two different ways will be demonstrated in the following tasks.

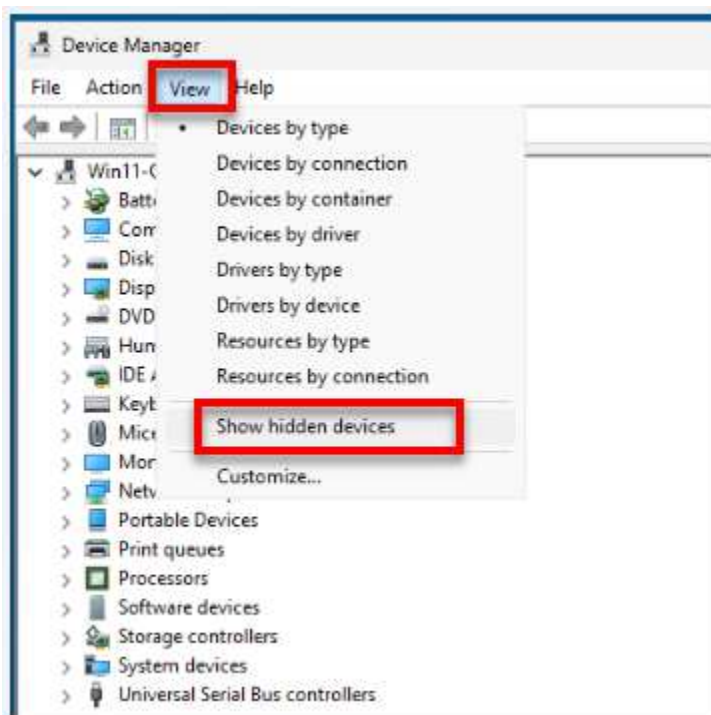
1. While on the **Win11-OS** System, click the **Search** Icon next to the Start Button, in the search window, type **Control Panel**. Click the **Control Panel App** Icon.



2. Make sure to select the drop-down arrow and change the *View by:* to **Small icons**, and click on the **Device Manager** icon.



3. Click **View** on the menu bar to see the other viewing options. The default view is *Device by type*. Some hardware devices don't appear by default, such as non-plug-and-play drivers, low-level system drivers included with *Windows*, or drivers installed by third-party software. Select **Show hidden devices** from the *View* tab menu to add hidden devices to the list.

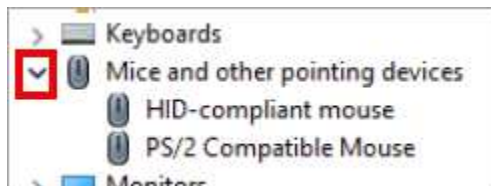


- Click on the **arrow** next to **Processors** in the main window of the default view list. The processors may differ from those shown in the screenshot below. This view details the *Processor* model and speed.

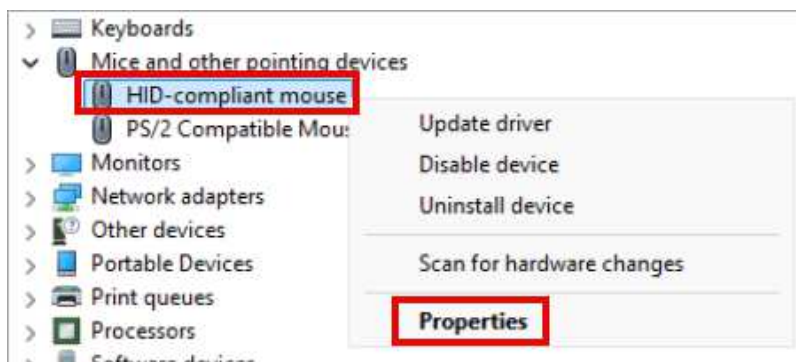


Clicking on the arrow next to any category and hardware component shows details. This allows you to view the exact model number or driver on this system hardware components installed on the system.

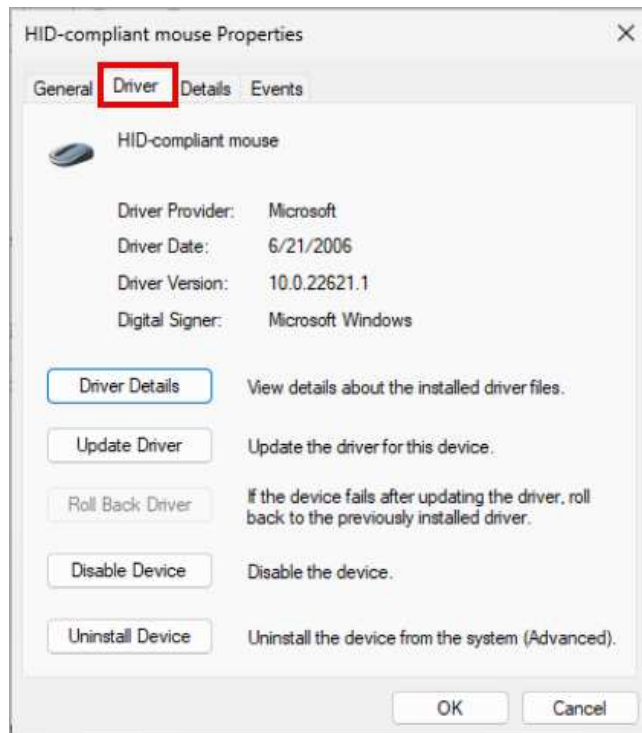
- Click on the **arrow** next to **Mice and other pointing devices**. This selection identifies the mouse hardware installed on this system.



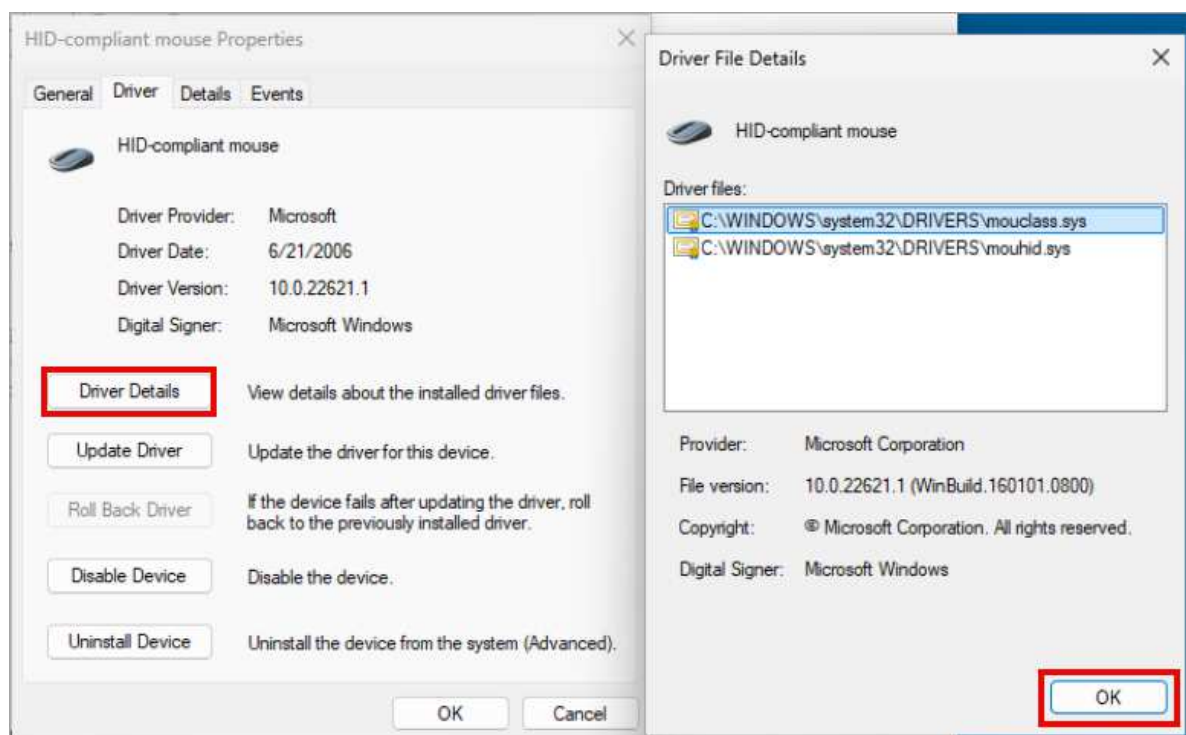
- Right-click on the **HID-compliant mouse** model listed, and a context menu opens with various options for managing the device. Click on **Properties** to view additional details.



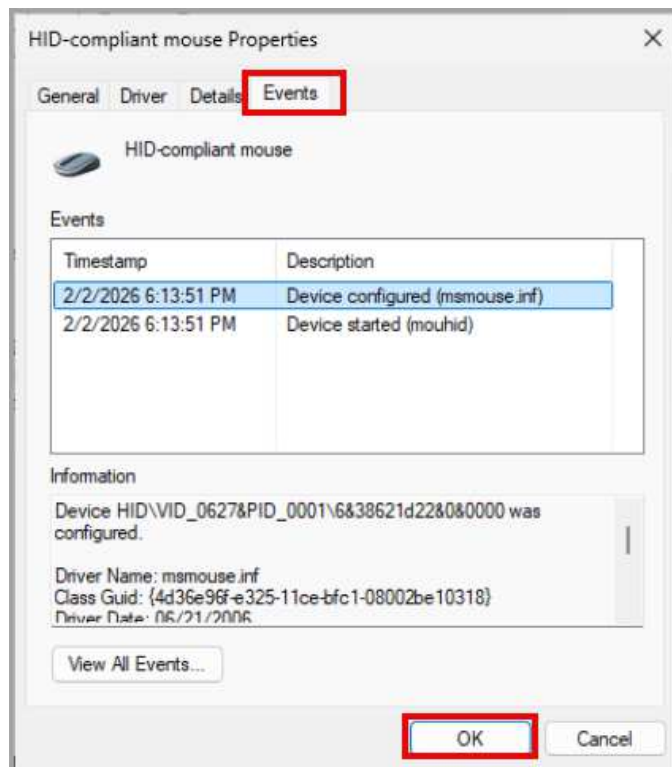
7. In *Windows 11*, there are four tabs shown: *General*, *Driver*, *Details*, *Events*, and *Resources*. Click on the **Driver** tab to view the pointing device details. This tab is very valuable for troubleshooting hardware issues. Review the buttons on the page to understand the options available in configuring the currently installed driver.



8. On the *Driver* tab, click on the **Driver Details** button. The Driver File Details page appears. Read the information on the page, and then click **OK** to close the window.



9. In the *Properties* window, click on the **Events** tab. This tab displays the history of *Windows* changes to the device. It can show when a device was installed, started, configured, updated, and more. This information may be useful for troubleshooting. Click **OK** to close the *Properties* page.



10. Close each remaining window by clicking the **X** in the top-right corner.

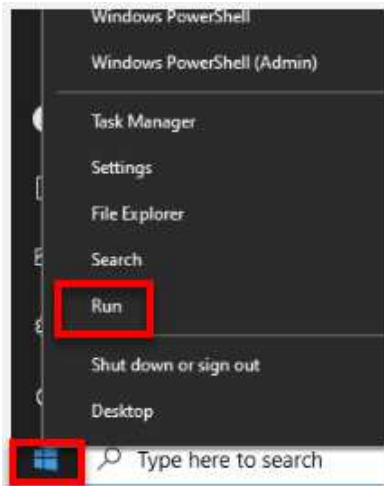


For comparison purposes:

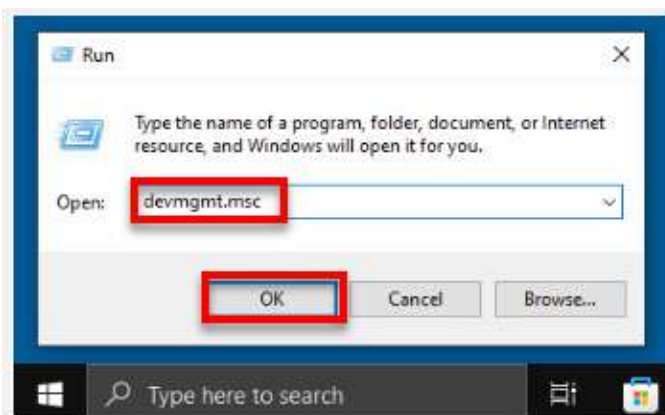
To access the *Device Manager* in *Windows 10* and *Windows 11*, right-click on the *Start* button, located on the taskbar in the lower-left corner of the screen. Then, click on *Device Manager* to open the program.

2.2 Accessing the Device Manager through the RUN Command

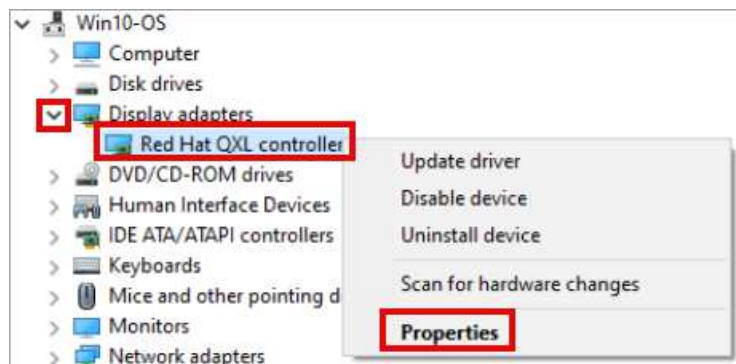
1. Switch to the **Win10-OS** system.
2. Right-click on the **Start** button located on the taskbar in the lower-left corner of the screen. The *Power User Menu* displays. Click on **Run** to open the *Run* command window.



3. In the *Run* command window, type `devmgmt.msc` and click **OK**. Notice that the *Device Manager* opens.



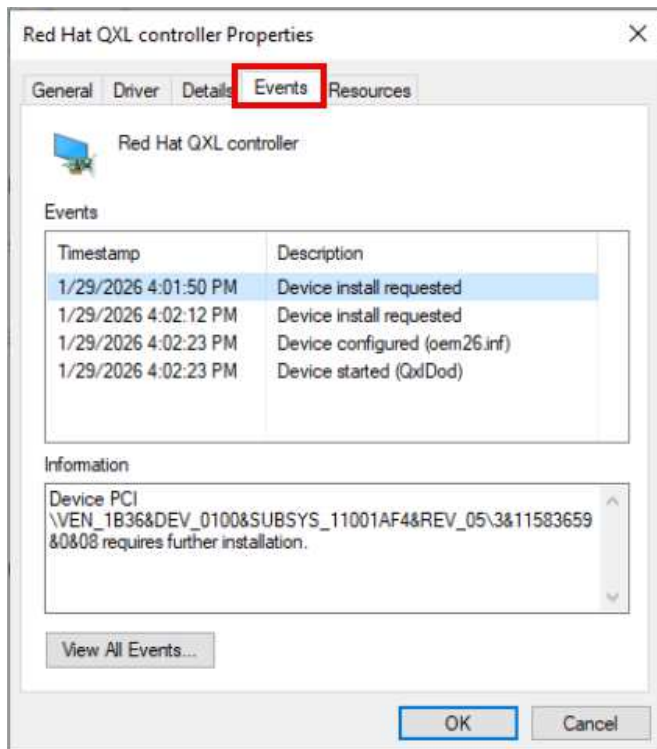
4. Click on the **arrow** next to **Display adapters** to expand the view and right-click the display adapter model listed, followed by clicking **Properties**.



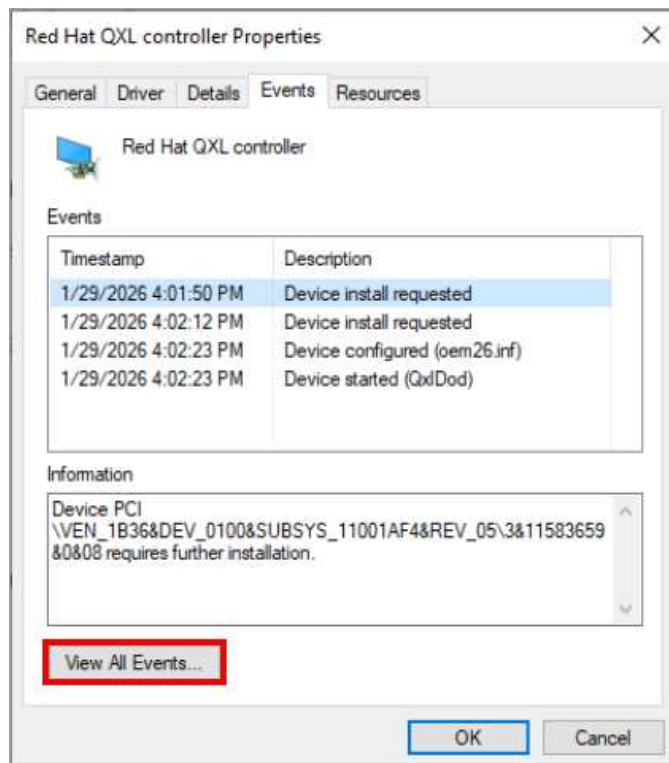


In *Windows 10*, there are five tabs shown: *General*, *Driver*, *Details*, *Events*, and *Resources*.

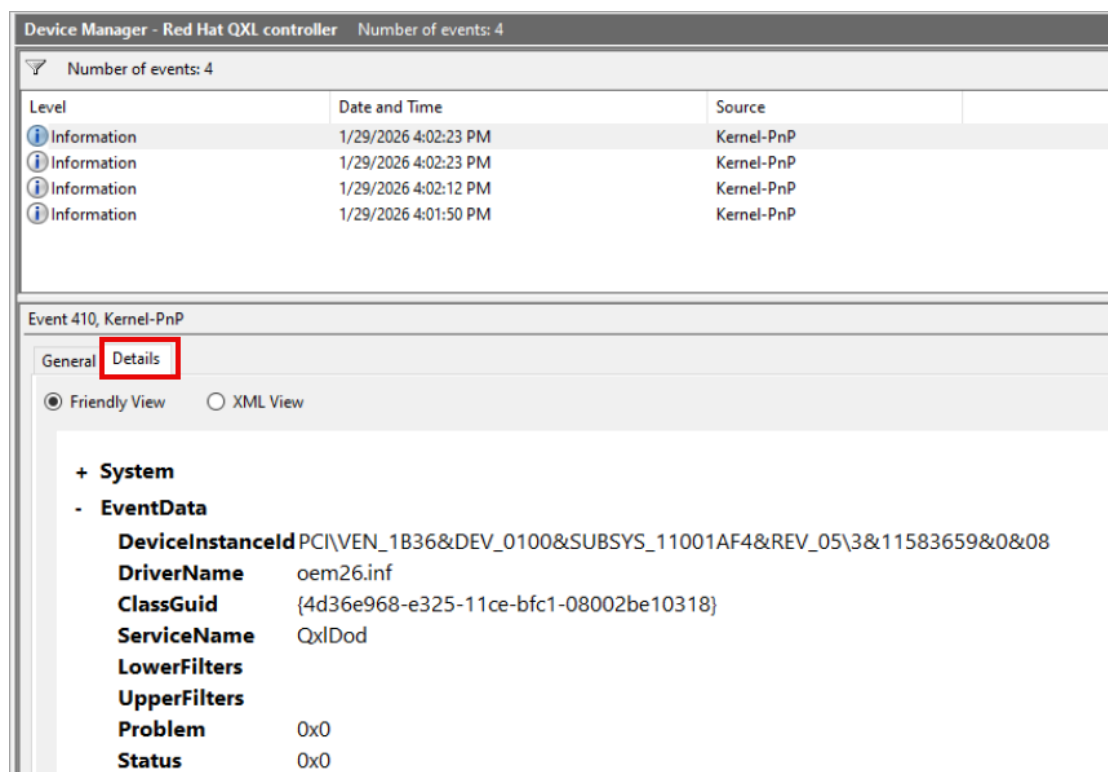
5. In the *Properties* window, click on the **Events** tab. This tab displays the history of *Windows* changes to the device. It can show when a device was installed, started, configured, updated, and more. This information may be useful for troubleshooting.



6. Click on **View All Events** to launch the *Event Viewer*, as this will display a detailed analysis of the events. You may receive an *Add Snap in Warning* before the *Event Viewer* window opens.



7. In the *Event Viewer* window, expand to full screen, click on the **Details** tab, and review the given information. Notice that more details are presented here regarding the display adapter. Close the *Event Viewer* when finished.



8. Click **OK** to close the *Properties* page.
9. The lab is now complete; you may end the reservation.